Statistical Mechanics of Turing Machines

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1 Turing Machines

I will define the basic objects I'm working with. I will treat a single Turing Machine as a dynamical system. In order to describe it, I need three variables: 1) The tape value at time step t and position x on the tape $V_t(x) \in [s] = \{0, 1, ..., s-1\}$. This a Turing machine that can write any symbol from a set of s symbols; 2) The position of the Turing Machine on this tape $x_t \in \mathbb{Z}$. For the moment, I'm assuming an infinite tape, but it might be necessary to consider periodic boundary conditions, in which case $x_t \in \mathbb{Z}_L$; 3) The state of the TM $q_t \in [n] := \{1, 2, ..., n\}$ for a $n-$state TM. To denote a Turing machine with n states over s symbols we write $\mathrm{TM}_n(s)$. I will focus mostly on $\mathrm{TM}_n(2)$, for which the tape consists of $\{0, 1\}$, but the TM can have arbitrarily many states.

The equations of motion of a single Turing machine are

$$x_{t+1} = x_t + M_{q_t}\left(V_t(x_t)\right) \tag{1}$$
$$V_{t+1}(x_t) = W_{q_t}\left(V_t(x_t)\right) \tag{2}$$
$$q_{t+1} = S_{q_t}\left(V_t(x_t)\right) \tag{3}$$

This is the most general formulation, and illustrates the fact that to define a TM we require three equal-sized matrices M, W, S. For each index q_t, the rows of these matrices define a function from the value of the tape that is read $V_t(x_t)$, to the corresponding response. M gives the next move, and so $M_q(b) \in \{-1, +1\}$. W gives the symbol that is written, so $W_q(b) \in [s]$. Finally, S tells the TM which state to transition to after being in state q_t and reading tape value $V_t(x_t)$. For this reason, $S_q(b) \in [n]$,

1.1 One-State Turing Machine: Langton's Ant

The first simple example we start with is Langton's ant, also called a turmite (presumably a portmanteau of turing and mite). Typically these are studied on a 2D grid, but I consider them on a one-dimensional tape. This model belongs to the class of $\mathrm{TM}_1(2)$, since there is only a single state and the behavior ultimately depends only on the current state of the tape. The rules are:

Read	Write	Move
0	1	$Left$
1	0	$Right$

For a single turmite, using the notation above, the update rules are

$$x^{\mu}_{t+1} = x^{\mu}_t + 2V_t(x^{\mu}_t) - 1 \tag{4}$$
$$V_{t+1}(x^{\mu}_t) = 1 - V_t(x^{\mu}_t) \tag{5}$$

There is a nicer representation in terms of the function $V_t(x)$, and the number of termites at x at time t:

$$n_t(x) = \sum_{\mu=1}^{N} \delta_{x\, x^{\mu}_t} \tag{6}$$

Without any explicit exclusion rule, there can be any number of turmites at a site, and their actions will all be in perfect agreement, so there is no risk of contradiction or interference. Note that with turing machine, one needs to add a rule about how conflicts are resolved, since adjacent TMs might be in different states, and consequently will want to take contradictory write actions on the same location.

Back to the turmite, we can find update rules for the tape value function and the density of turmites:

$$V_{t+1}(x) = V_t(x) + \Theta(n_t(x))\,(1 - 2V_t(x)) \tag{7}$$
$$n_{t+1}(x) = (1 - V_t(x+1))\,n_t(x+1) + V_t(x-1)n_t(x-1) \tag{8}$$

where $\Theta(n_t(x)) = 1$ for $n_t(x) > 0$ and 0 otherwise. The total number of turmites never changes, and so $N = \sum_x n_t(x)$ is conserved. This implies at least a continuity equation. Let's define the chiral current flux. In the right direction, let $J_R(x-1, x)$ denote the total flux from $x-1$ to x. This just

$$J_R(x-1, x) = V_t(x-1)n_t(x-1) - (1 - V_t(x))n_t(x) \tag{9}$$

Similarity, the leftward flux from $x+1$ to x is

$$J_L(x+1, x) = (1 - V_t(x+1))n_t(x+1) - V_t(x)n_t(x) = -J_R(x, x+1) \tag{10}$$

Using these fluxes, I get

$$n_{t+1}(x) - n_t(x) = J_R(x-1, x) + J_L(x+1, x) = J_R(x-1, x) - J_R(x, x+1) \tag{11}$$

which is the discrete version of the continuity equation. There is a different presentation of this in the appendix.

1.1.1 Turmitic Solitons

The simulations suggest looking for solitonic solutions which are ballistic. Let us assume that $n_t(x) = N\delta_{x0}$ for a doubly infinite tape, and $V_t(x) = 1$. Then

$$V_{t+1}(0) = 0, \quad n_{t+1}(1) = N, \quad n_{t+1}(0) = 0 \tag{12}$$
$$V_{t+2}(1) = 0, \quad n_{t+2}(2) = N, \quad n_{t+2}(1) = 0 \tag{13}$$

In general, let's just assume initial $t = 0$. Then

$$V_t(x) = \sum_{x'<t} \delta_{x,x'}, \quad n_t(x) = N\delta_{x,t} \tag{14}$$

Which is clearly a soliton moving at velocity $v = 1$. Now let's say that $V_0(x) = 0$ for $x > L$. Then this soliton will move until it hits $x = L$, then it will be reflected and start moving at $v = 1$ in the opposite direction.

Another thing to notice: If all turmites occupy the same point, there is no mechanism for them to disperse. Therefore, unless the individual turmites never collide, the most likely asymptotic configuration is for all of the turmites to coalesce into one multi-cellular monster, moving as a soliton throughout the tape.

1.2 Two-State Turing Machines: Multi-species turmites

Now let's try to extend the analysis of Langton's ant to simple 2 state turing machines.

2 Multi-agent systems

The turing machines defined above are very elementary agents. They are stateful, and act deterministically on their environment, which is the tape. The action taken by the TM depends on the current bit E_t and the current state S_t, which can be encapsulated via

$$A_{t+1} \sim P(A|S_t, E_t) \tag{15}$$

The actions just consist of transformations of E_t. Notice that while the TM is stateful, it is Markovian. The configuration (E_t, S_t) only depends on the previous time step. Is there a nontrivial yet tractable way to include memory in this system? For instance

AI scratchpad

A Prior Works

The systems considered here consist of mobile finite-state machines that read and modify a shared tape. Their collective behavior connects several existing research programs: cellular automata, interacting turmites, algorithmic chemistry, deterministic walks in a writable environment, and nonequilibrium statistical mechanics. The distinction between these connections matters. Some references study nearly the same microscopic architecture; others supply observables or solution methods whose assumptions must be checked for the present dynamics.

A.1 Cellular automata, mobile automata, and rule-space surveys

Wolfram's statistical-mechanical study of cellular automata [1] is a methodological precedent for sampling simple deterministic rules, evolving random initial conditions, and comparing the resulting organization. It demonstrates that local binary rules can generate structured statistical behavior without an equilibrium energy function. In a cellular automaton, the sites update through a prescribed spatial rule; in the present simulations, the moving heads determine where updates occur. Their positions therefore form an additional dynamical field.

A New Kind of Science [2] extends systematic experimentation to Turing machines and mobile automata, including generalized mobile automata with multiple active cells. These examples motivate the spacetime zoos and comparisons between blank and random tapes. The associated 2003 collection of open problems [3] calls for systematic studies of simple Turing machines, nonblank initial conditions, and the relation between active-cell density and behavioral complexity. Those historical questions closely match the present program, but their appearance in that collection does not establish their current research status. Wolfram's later multiway Turing machines [4] concern branching computational histories under nondeterministic rules. Multiple branches are distinct from a population of heads simultaneously modifying a single tape.

A.2 Interacting turmites and shared-tape agents

Belgacem and Fatès [5] provide a direct spatial precedent by studying multiple Langton turmites on a shared binary lattice. They compare synchronous updates, a fixed sequential order, and random sequential orders, together with different policies for spatial conflicts. Their experiments show that collective patterns depend on the scheduling and interaction conventions, even when the individual agent rule stays fixed. This supports treating update order and collision resolution as model parameters in our simulations. Their model is two-dimensional and includes agent orientation and movement-conflict policies, so its collective structures do not directly determine the behavior of the one-dimensional rules studied below.

Polhill and collaborators [6] explicitly consider a population of Turing machines reading and writing the same tape. Their thought experiment studies prediction when transition tables, initial head states and positions, and execution schedules are hidden. It establishes a close architectural precedent, while addressing inference and computability rather than deriving stationary measures or transport laws. In particular, its discussion of noncomputable schedules should be distinguished from the reproducible, seeded random schedules used in numerical experiments here.

A.3 Algorithmic chemistry and Fontana's Turing gas

Fontana's algorithmic chemistry [7] treats an ensemble of symbolic functions as interacting objects. In this "Turing gas," expressions derived from the lambda calculus act on one another to generate new expressions, and the population develops cooperative organizations. This supplies an early precedent for studying collective behavior in a population of programs.

Fontana and Buss [8] subsequently analyze the emergence of reproducing and self-maintaining organizations in this abstract chemistry. Their work suggests examining which interaction patterns persist across different initial populations. The correspondence to our model is conceptual: their reacting objects are themselves programs, and reactions change the program population. Here the sampled transition tables remain fixed and the shared spatial tape carries the evolving record of interactions. Thus algorithmic chemistry motivates questions about collective organization, but its reaction rules cannot be imported as the dynamics of the present gas.

A.4 Eulerian walkers and rotor-router theory

The symbol-flipping one-state machine has a particularly close connection to rotor walks: successive occupied visits alternate the departure direction at a site. Priezzhev and collaborators [9] study walkers that modify local arrows and, on open graphs, are replaced after absorption. Their particle addition operators connect the stationary problem to the Abelian sandpile model. Holroyd and collaborators [10] give a systematic treatment of rotor routing, its Abelian properties, and the relation between recurrent rotor configurations and rooted spanning trees.

These results apply directly to the isolated-particle excursions used in the dilute boundary-driven calculation in Appendix B.3, after matching the convention for whether a rotor records the last or next departure. In particular, the recurrent empty tapes 1^k0^{L-k} are the one-dimensional version of acyclic rotor configurations with absorbing boundaries. The dilute calculation should therefore be understood in the context of established rotor-router theory, rather than as an unrelated mechanism.

Klasing and collaborators [11] analyze several agents sharing a rotor environment on a ring, obtaining rigorous cover-time and long-time return-time results. A decisive microscopic difference is that their rotor advances once per departing

agent: two agents at one site can leave in opposite directions. In our synchronous symbol-flipping model, the tape flips once per *occupied site*, and all particles in a stack depart together. Stacks that meet at one site merge permanently. The published multi-agent bounds and Abelian arguments therefore do not automatically extend to this coalescing dynamics. They identify useful tools and solvable sectors, while leaving a distinct problem at finite stack density.

A.5 Quantifying tape structure and relaxation

Computational mechanics provides observables that distinguish visible organization from simple changes in symbol frequency. Shalizi and collaborators [12] develop local sensitivity and statistical-complexity filters for detecting coherent structures in spatiotemporal systems. Applied to cellular automata, these reveal domains and localized structures without assuming their shapes in advance. For the present spacetime plots, such filters suggest a way to quantify traveling structures and collision regions, supplementing head trajectories and tape density.

Feldman, McTague, and Crutchfield [13] compare processes using complexity–entropy diagrams, combining randomness with stored structure. This is relevant because a tape with equal numbers of zeros and ones can still be periodic or strongly correlated. Useful measurements here would include block entropies, entropy rates, excess entropy, spatial correlations, and correlations between head states and local symbols. Estimating them requires specifying the ensemble and checking stationarity; pooling different machine rules or different ages of a relaxing system can otherwise obscure the distinction between within-run fluctuations and variation across runs. Convergence of these observables would provide evidence of statistical relaxation, but would not by itself identify a Gibbs ensemble or a temperature.

A.6 Boundary-driven states and thermodynamic costs

The review by Buča, Klobas, and Prosen [14] describes exact nonequilibrium stationary states of the reversible Rule-54 cellular automaton with stochastic boundaries, including patch and matrix-product representations, relaxation modes, and current fluctuations. It supplies a concrete strategy for the boundary-driven attempt below: specify the reservoirs, construct the Markov operator, solve small systems, and search for a representation whose local algebra verifies stationarity. Its exact formulas depend on that model's microscopic structure. Our synchronous coalescence rule does not inherit those formulas merely because it also admits a cellular-automaton description. The transferable object is the stationary-state problem and the methods used to attack it.

Strasberg and collaborators [15] address a different thermodynamic question by constructing stochastic physical implementations of Turing machines with consistent entropy production. Their results concern the cost of performing computation. A statistical ensemble of abstract shared-tape machines does not yet specify the energy levels, physical transition rates, or reservoirs needed to assign

heat and work. For this reason, tape Shannon entropy and state-distribution entropy should initially be treated as information-theoretic observables, not identified with physical entropy production.

A.7 Scope of the present problem

Together, these references establish precedents for computational populations, agents coupled through writable media, and statistical descriptions of deterministic local dynamics. Within the works reviewed here, we have not identified a general stationary or hydrodynamic theory covering arbitrary randomly sampled transition tables under the present shared-tape collision rules; this is a statement about the scope of this review, not a claim of priority.

A useful formulation separates randomness in the initial tape and head states from randomness in the transition tables. Sampling a table once and retaining it defines quenched algorithmic disorder; resampling it during evolution would define a different process. The closed-system limit should specify particle number N, tape length L, and density N/L, while an open-system limit must also fix how reservoir rates scale with L. The one-state symbol-flipping model below isolates a tractable part of this broader program: it connects directly to rotor walks in the single-stack sector and introduces irreversible coalescence when several stacks interact synchronously.

B Notes on Langton's Ant

The rule considered above is

$$0 \longrightarrow (1, L), \qquad 1 \longrightarrow (0, R). \tag{16}$$

It is useful to distinguish the number of turmites at a site from the Boolean event that the site is occupied. Define

$$I_t(x) = \mathbf{1}_{\{n_t(x)>0\}}. \tag{17}$$

Thus, throughout this section, $\Theta(n)$ is understood to equal zero at $n = 0$ and one for $n > 0$.

Under synchronous updating, all turmites first read the tape at time t, then propose their writes and moves. Since all turmites at a given site read the same symbol, their write proposals agree. The exact tape update is therefore

$$V_{t+1}(x) = V_t(x) + I_t(x)\,[1 - 2V_t(x)] \tag{18}$$

$$= V_t(x) \oplus I_t(x), \tag{19}$$

where $\oplus$ denotes addition modulo two.

A turmite at $x+1$ enters x if it reads zero and moves left, whereas a turmite at $x-1$ enters x if it reads one and moves right. Consequently, the exact number update is

$$\boxed{n_{t+1}(x) = [1 - V_t(x+1)]\,n_t(x+1) + V_t(x-1)n_t(x-1).} \tag{20}$$

This agrees with the number update in the main text. If the equation is instead written for the increment $n_{t+1}(x) - n_t(x)$, its right-hand side must include the additional loss term $-n_t(x)$.

To display number conservation, define the rightward current across the bond $x + \frac{1}{2}$ by

$$J_t\left(x + \tfrac{1}{2}\right) = V_t(x) n_t(x) - \left[1 - V_t(x+1)\right] n_t(x+1). \tag{21}$$

The number equation then takes the continuity form

$$n_{t+1}(x) - n_t(x) = J_t\left(x - \tfrac{1}{2}\right) - J_t\left(x + \tfrac{1}{2}\right). \tag{22}$$

On an infinite tape with suitable decay, or on a periodic tape with all site indices understood modulo L, this implies

$$\sum_x n_{t+1}(x) = \sum_x n_t(x) = N. \tag{23}$$

B.1 Rotor representation

Define a binary rotor field

$$\sigma_t(x) = 2V_t(x) - 1 \in \{-1, +1\}, \tag{24}$$

where -1 sends a stack left and $+1$ sends it right. The coupled equations become

$$\sigma_{t+1}(x) = (-1)^{I_t(x)} \sigma_t(x), \tag{25}$$

$$n_{t+1}(x) = \frac{1 - \sigma_t(x+1)}{2} n_t(x+1) + \frac{1 + \sigma_t(x-1)}{2} n_t(x-1). \tag{26}$$

The first equation says that an occupied site reverses its rotor exactly once, independently of the number of turmites in the stack.

The tape and the support of the turmite distribution form a closed Boolean system. In particular,

$$\boxed{I_{t+1}(x) = \left[I_t(x-1) \wedge V_t(x-1)\right] \vee \left[I_t(x+1) \wedge \neg V_t(x+1)\right].} \tag{27}$$

Together with $V_{t+1} = V_t \oplus I_t$, this is a deterministic, nearest-neighbor four-state cellular automaton for (V_t, I_t). The integer masses $n_t(x)$ can be transported after the Boolean dynamics have been solved.

Every occupied site sends its entire stack to one neighboring site. Two stacks that arrive at the same site merge and can never split again. Hence the number of occupied sites

$$K_t = \sum_x I_t(x) \tag{28}$$

is nonincreasing:

$$K_{t+1} \le K_t. \tag{29}$$

This monotonic coalescence property is a useful simplification even though the conserved mass N need not decrease.

B.2 Odometer representation and a route to a solution

Introduce the site odometer

$$u_t(x) = \sum_{\tau=0}^{t-1} I_\tau(x), \tag{30}$$

which counts the number of time steps at which site x has been occupied. The tape can then be eliminated in favor of its initial condition:

$$\sigma_t(x) = \sigma_0(x)(-1)^{u_t(x)}. \tag{31}$$

For a single stack at position X_t, this gives the closed rotor-walk form

$$X_{t+1} = X_t + \sigma_0(X_t)(-1)^{u_t(X_t)}. \tag{32}$$

A possible route toward solving the many-turmite system is the following.

1. Solve the single-stack rotor walk in terms of its odometer, visited interval, and return times.

2. Classify two-stack scattering and determine the conditions for coalescence. On an even ring, each step changes position parity, so two stacks can occupy the same site at the same time only if their initial positions have the same parity.

3. Analyze the four-state cellular automaton (V, I) on small periodic rings by constructing its complete transition graph. This gives exact attractors, periods, traveling waves, and basin sizes without retaining the stack masses.

4. Use the fact that a deterministic system on a finite ring must eventually become periodic. A traveling orbit that advances by d sites in P time steps has the rational velocity

   $$v = \frac{d}{P}. \tag{33}$$

   This provides a possible explanation for the recurring rational velocities observed numerically.

5. Once the support dynamics and its coalescence events are known, propagate the conserved integer masses along those trajectories to recover $n_t(x)$ and the mass-weighted velocity distribution.

The other symbol-flipping one-state machine, $0 \longrightarrow (1, R)$ and $1 \longrightarrow (0, L)$, is obtained from the system above by the spatial reflection $x \mapsto -x$. Solving one rule therefore solves its mirror as well.

B.3 An attempt at a boundary-driven solution

The boundary-driven strategy is to keep the deterministic bulk rule and introduce randomness only through particle reservoirs. One then solves for a stationary probability distribution and its currents. The review by Buča, Klobas, and Prosen [14] illustrates this strategy using patch and matrix-product stationary states. Here we apply the general strategy to the symbol-flipping model itself. The results below include exact stationary identities, an existence and uniqueness argument, and a solution to leading order in the injection rates; an arbitrary-density closed-form stationary distribution remains to be found.

Choice of reservoirs and update convention. Take an interval $x = 1, \dots, L$, with $L \geq 2$. Particles leaving either end are absorbed. At each time step, independent variables

$$b_t^{\mathrm{L}} \sim \mathrm{Bernoulli}(\alpha), \qquad b_t^{\mathrm{R}} \sim \mathrm{Bernoulli}(\beta) \tag{34}$$

inject one particle at the left and right endpoints, respectively. These variables are independent of the configuration and of previous injections. First the old particles read, flip, and move synchronously; the new particles are then added to the endpoints and first act on the next step. The reservoirs do not reset tape symbols. Other reservoir conventions define different open models and need not have the stationary state derived here.

With time subscripts suppressed on the right-hand sides, the boundary rules are

$$n_{t+1}(1) = (1 - V_2)n_2 + b_t^{\mathrm{L}}, \tag{35}$$

$$n_{t+1}(L) = V_{L-1}n_{L-1} + b_t^{\mathrm{R}}, \tag{36}$$

with Eq. (20) in the interior and $V_{t+1}(x) = V_x \oplus I_x$ everywhere. Injection into an already arriving stack adds mass without creating a second stack. In particular,

$$N_{t+1} - N_t = b_t^{\mathrm{L}} + b_t^{\mathrm{R}} - (1 - V_1)n_1 - V_L n_L. \tag{37}$$

A finite Markov problem for tape and support. Define the outgoing stack indicators

$$r_x = I_x V_x, \qquad \ell_x = I_x(1 - V_x). \tag{38}$$

The closed support dynamics are

$$I_{t+1}(1) = b_t^{\mathrm{L}} \vee \ell_2, \tag{39}$$

$$I_{t+1}(x) = r_{x-1} \vee \ell_{x+1}, \quad 2 \leq x \leq L - 1, \tag{40}$$

$$I_{t+1}(L) = r_{L-1} \vee b_t^{\mathrm{R}}. \tag{41}$$

Thus $s = (V, I)$ has 4^L possible values, although the full state (V, n) has unbounded integer masses. Let $F_{bc}(s)$ denote the Boolean update with boundary choices (b, c) and let

$$w_{bc} = \alpha^b(1 - \alpha)^{1-b}\beta^c(1 - \beta)^{1-c}. \tag{42}$$

The column-stochastic transition matrix and stationary equation are

$$P(s' \mid s) = \sum_{b,c=0}^{1} w_{bc} \mathbf{1}_{\{s'=F_{bc}(s)\}}, \qquad P\pi = \pi, \qquad \sum_{s} \pi(s) = 1. \tag{43}$$

Different injection events can produce the same Boolean successor, so their probabilities must be added. This is an exact finite-size formulation; its exponential state count is the practical limitation.

Why the finite open system has a unique steady state. There is a useful decreasing potential when injection is switched off. Set

$$h_x = x(L+1-x), \qquad a_x = L+1-2x, \tag{44}$$

and define

$$\mathcal{Q}(V,I) = \sum_{x=1}^{L} h_x I_x + \sum_{x=1}^{L} \big[a_x V_x - \min(0, a_x)\big]. \tag{45}$$

Let $C_x = r_{x-1}\ell_{x+1}$, setting the incoming indicators outside the interval to zero. With no injection, direct substitution gives

$$\boxed{\mathcal{Q}_{t+1} - \mathcal{Q}_t = -K_t - \sum_{x=1}^{L} h_x C_x.} \tag{46}$$

Indeed, a departure in direction $\sigma_x = 2V_x - 1$ changes its h weight by $h_{x+\sigma_x} - h_x = -1 + a_x\sigma_x$, while the tape contribution changes by $-a_x\sigma_x$. Coalescence removes one additional h_x weight at each double arrival. Absorption is included by $h_0 = h_{L+1} = 0$. Since

$$0 \le \mathcal{Q} \le B_L, \qquad B_L = \frac{L(L+1)(L+2)}{6} + \left\lfloor \frac{L^2}{2} \right\rfloor, \tag{47}$$

every configuration empties within B_L consecutive steps without injection, independently of its stack masses.

Emptying does not reset the tape, but a common empty configuration is accessible. Inject one particle from the left into an empty interval and wait for it to exit. If the first zero is at j, the particle travels to j, turns, and exits on the left; its final tape has its first j symbols equal to one and its remaining symbols unchanged. If all symbols were one, it exits on the right and leaves an all-zero tape. At most $L+1$ such isolated injections therefore reach the empty all-zero configuration, denoted $s_\star$. Each excursion takes at most $2L-1$ moves.

For $0 < \alpha, \beta < 1$, all the required injection and no-injection events have positive probability. Every Boolean state can reach $s_\star$, which also has a no-injection self-loop. Hence there is exactly one recurrent class, it is aperiodic, and π is unique. This argument does not require irreducibility on all 4^L states.

The same construction reaches $(V, n) = (0, 0)$ in a uniformly bounded number of steps, with a positive probability bounded below for fixed (L, α, β), regardless of the initial masses. Repeated blocks thus give a geometric bound on the reset time. Since at most two particles enter per step, the full process also has a unique stationary distribution with finite mass moments. The bounds deteriorate with L and do not establish a thermodynamic-limit steady state.

Exact stationary identities and the first closure problem. Write $\langle\cdot\rangle$ for stationary expectation and $\kappa_x = \langle I_x\rangle$ for the stack density. Stationarity of the tape gives a particularly simple identity:

$$0 = \langle V_{t+1}(x) - V_t(x)\rangle = \langle I_x(1 - 2V_x)\rangle, \qquad \boxed{\langle r_x\rangle = \langle \ell_x\rangle = \frac{\kappa_x}{2}.} \tag{48}$$

Thus the symbol seen at an occupied site is unbiased in the stationary measure. This does not imply $\langle V_x\rangle = 1/2$: the tape at empty sites can be strongly polarized.

Using $a \vee b = a + b - ab$, the bulk support balance becomes

$$\boxed{\frac{\kappa_{x-1} - 2\kappa_x + \kappa_{x+1}}{2} = c_x, \qquad c_x = \langle r_{x-1}\ell_{x+1}\rangle \geq 0.} \tag{49}$$

The mean stack profile is therefore discretely convex. Its rightward current is $j_{x+1/2} = (\kappa_x - \kappa_{x+1})/2$ and satisfies $j_{x-1/2} - j_{x+1/2} = c_x$. Independence of each fresh boundary draw gives the exact boundary conditions

$$\kappa_1 = \alpha + (1 - \alpha)\frac{\kappa_2}{2}, \qquad \kappa_L = \beta + (1 - \beta)\frac{\kappa_{L-1}}{2}. \tag{50}$$

For $L = 2$, these close without any approximation:

$$\kappa_1 = \frac{2(2\alpha + \beta - \alpha\beta)}{3 + \alpha + \beta - \alpha\beta}, \qquad \kappa_2 = \frac{2(\alpha + 2\beta - \alpha\beta)}{3 + \alpha + \beta - \alpha\beta}. \tag{51}$$

For larger systems, the unknown collision correlations c_x prevent a one-point closure. The replacement $c_x \approx \kappa_{x-1}\kappa_{x+1}/4$ is a possible mean-field approximation, not an exact consequence of the rotor balance.

Mass has a different balance: its stationary current J is constant across all bonds, with

$$J = \langle V_x n_x - (1 - V_{x+1})n_{x+1}\rangle, \tag{52}$$

$$J = \alpha - \langle(1 - V_1)n_1\rangle = \langle V_L n_L\rangle - \beta. \tag{53}$$

Equal left and right *stack* departure rates do not imply equal *mass* departure rates: stack size can correlate with direction.

Exact finite-size mass observables without a mass cutoff. There is nevertheless a finite linear closure for first mass moments conditioned on the entire Boolean state. Let $T(V)$ route each column x to row $x + 2V_x - 1$, deleting it if the destination lies outside the interval. For injection event $e = (b, c)$, set $g_e = b\,\mathbf{e}_1 + c\,\mathbf{e}_L$. The full mass update is $n' = T(V)n + g_e$. Define the unnormalized vector

$$m_x(s) = \mathbb{E}\big[n_x \mathbf{1}_{\{(V,I)=s\}}\big]\,. \tag{54}$$

After solving Eq. (43), stationarity requires

$$m(s') = \sum_{s,e} w_e \mathbf{1}_{\{s'=F_e(s)\}}\big[T(V(s))m(s) + g_e\pi(s)\big]. \tag{55}$$

This is a linear system for at most $L4^L$ unknowns, with $m_x(s) = 0$ if $I_x(s) = 0$. Its homogeneous evolution transports only previously present mass, which is erased by a sufficiently long no-injection block. On these support-compatible coordinates its spectral radius is less than one, so the linear system has a unique solution. Then

$$\rho_x = \sum_s m_x(s), \qquad J = \sum_s \left[V_x(s)m_x(s) - (1 - V_{x+1}(s))m_{x+1}(s)\right] \tag{56}$$

give the exact finite-size mass density and current. This bypasses a cutoff on stack size, although the Boolean state count remains exponential.

A controlled solution for rare injection. Set $\alpha = \epsilon a$, $\beta = \epsilon b$, with fixed $a, b > 0$, and take $\epsilon \to 0$ at fixed L. Injections are then separated by isolated single-particle excursions with probability tending to one. A naive expansion about a fixed empty tape would fail: at zero injection, all 2^L empty tapes are absorbing. The drive selects a measure on this degenerate set, which must be determined first.

The recurrent empty tapes under isolated boundary injections are

$$E_k = \underbrace{11\cdots1}_{k}\underbrace{00\cdots0}_{L-k}, \qquad k = 0, \ldots, L. \tag{57}$$

A left injection maps E_k to E_{k+1}, and a right injection maps it to E_{k-1}, with indices modulo $L+1$. This set is closed, and the left-injection construction above takes every other empty tape into it. On the slow time scale, the interface index has generator

$$(\mathcal{G}f)(k) = a[f(k+1) - f(k)] + b[f(k-1) - f(k)]. \tag{58}$$

Its stationary distribution is uniform, even when $a \neq b$. Because the waiting time for injection is independent of k, the physical-time stationary measure has the same leading weights:

$$\boxed{\pi(V = E_k, I = 0) = \frac{1}{L+1} + O(\epsilon).} \tag{59}$$

The tape is consequently correlated and polarized even at arbitrarily weak drive. For example,

$$\langle V_x \rangle = \frac{L+1-x}{L+1} + O(\epsilon), \tag{60}$$

$$\mathrm{Cov}(V_x, V_y) = \frac{x(L+1-y)}{(L+1)^2} + O(\epsilon), \qquad x \le y. \tag{61}$$

The same excursion picture gives the particle densities. Averaged uniformly over E_k, a left-injected particle visits site x an average $2(L+1-x)/(L+1)$ times before absorption. A right-injected particle has mean visit count $2x/(L+1)$. Multiplying by the injection rates yields

$$\boxed{\rho_x = \kappa_x + O(\epsilon^2) = \frac{2\big[\alpha(L+1-x) + \beta x\big]}{L+1} + O(\epsilon^2).} \tag{62}$$

The mean residence time of an isolated injected particle is L steps, and hence $\langle N \rangle = L(\alpha+\beta) + O(\epsilon^2)$. A left-injected particle exits on the right only from E_L; a right-injected particle exits on the left only from E_0. Therefore

$$\boxed{J = \frac{\alpha-\beta}{L+1} + O(\epsilon^2).} \tag{63}$$

These are fixed-L asymptotic statements. The coefficients hidden in the errors can grow with L; in particular, overlap becomes relevant when the injection rate times the excursion duration is no longer small. The $1/(L+1)$ conductance is an exact leading dilute-limit result, not yet a proof of diffusive transport at finite density.

A matrix-product starting point and the remaining problem. The leading empty-tape distribution already has an exact bond-dimension-two matrix-product representation. In the physical basis (V, I), take

$$A_{(1,0)} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad A_{(0,0)} = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \qquad A_{(v,1)} = 0, \tag{64}$$

with $\langle l| = (1,0)$ and $|r\rangle = (1,1)^{\mathsf{T}}$. Then

$$\pi^{(0)}(s) = \frac{1}{L+1} \langle l | A_{(V_1,I_1)} \cdots A_{(V_L,I_L)} | r \rangle \tag{65}$$

assigns unit unnormalized weight precisely to the $L+1$ tapes E_k with empty support. This supplies an explicit starting point for a perturbative matrix-product calculation rather than an unspecified ansatz.

At nonzero density, excursions overlap, stacks merge, and tape memory couples successive events. A next step is to calculate the $O(\epsilon^2)$ collision corrections using two injections, including both same-boundary and opposite-boundary

events. In parallel, the finite-size stationary vectors from Eq. (43) can test whether a small matrix-product representation survives at finite drive. An ansatz must satisfy the actual local update and reservoir equations; no fixed bond dimension or Rule-54 cancellation algebra has been established for this model. Mass observables can then be obtained from Eq. (55) even if the full mass distribution is not explicitly known.

Checks on the calculation. The no-injection potential identity was verified on all 4^L Boolean configurations for $L = 2, \ldots, 8$. Their maximal clearing times were $3, 7, 12, 18, 26, 37, 47$ steps, respectively, all below B_L. The isolated-injection tape maps were checked for every empty tape through $L = 10$. Numerical stationary solutions of the exact finite transition matrix at $(\alpha, \beta) = (0.37, 0.61)$ satisfied the rotor, boundary, and coalescence identities to better than 10^{-11} for $L = 2, \ldots, 7$; the conditional mass equations gave spatially constant mass currents through $L = 6$. At $L = 3$, with $(a, b) = (0.37, 0.61)$ and $\epsilon = 0.02, 0.01, 0.005$, the discrepancies in Eqs. (62) and (63) scaled as $O(\epsilon^2)$. These checks support the derivations but do not replace an arbitrary-density solution.

C Spacetime zoo

The thirteen grids contain twelve ensembles of 100 realizations and all sixteen one-state rules. All machines have $s = 2$ symbols. Following Section 1, n counts internal states, $q_t^{(i)} \in [n]$ and $x_t^{(i)}$ are head i's state and position, and M_q, W_q, S_q are its move, write, and next-state functions. There are N heads and the observation duration is T.

Reading the plots. Within every panel, x increases to the right and t increases upward from 0 to T. Dark and light pixels represent $V_t(x) = 0$ and 1. Red overlays indicate $n_t(x) = 1$ and yellow overlays indicate $n_t(x) > 1$, where $n_t(x) = \sum_{i=1}^{N} \mathbf{1}_{\{x_t^{(i)} = x\}}$ counts heads. Colors do not encode $q_t^{(i)}$; co-location need not imply permanent coalescence for machines with internal states. Snapshots were recorded every five steps and further downsampled for these overviews, so thin trajectories can be omitted. Zooming reveals the original panel labels.

Ensembles and initial conditions. For a randomly sampled $\mathrm{TM}_n(2)$, each table entry independently chooses $W_q(b) \in \{0, 1\}$, $M_q(b) \in \{-1, +1\}$, and $S_q(b) \in [n]$ uniformly. "Identical" means that all N heads share one sampled triple (M, W, S); "independent" means that each head has its own sampled triple. The tables remain fixed during a run. In every collection,

$$x_0^{(i)} = 5\big(i - 1 - \lfloor N/2 \rfloor\big), \qquad i = 1, \ldots, N, \tag{66}$$

with positions reduced modulo L on a periodic tape. Thus the spacing is five sites; for $N = 1000$ and $L = 10000$ the heads initially occupy only part of the ring. Random tapes have independent Bernoulli-(p_1) symbols with $p_1 = 1/2$; blank tapes have $V_0(x) = 0$. Fixed initial states mean $q_0^{(i)} = 1$; random states are independent uniform draws from $[n]$.

Updates, boundaries, and panel labels. Synchronous steps read the old tape and use majority writes (ties preserve the old symbol), then move all heads. In the asynchronous collection, one time unit is a sweep in which every head acts once in a fresh uniformly random order; each action immediately reads, writes, changes state, and moves. No simultaneous-write vote is used there. "Open" in the saved plots means an effectively unbounded tape $x \in \mathbb{Z}$, with a finite display window: heads leaving that window remain in the simulation. Periodic panels show all $x \in \mathbb{Z}_L$. None of these zoos injects or absorbs particles. In the original headings, `S` means our n, `W` is the display width, `dx` is the initial spacing, and `r` is the recording interval; these labels do not denote the functions S_q and W_q. Panels are ordered left to right and then top to bottom, by seed or one-state rule ID. Matching seeds across different ensemble sizes or tape geometries do not in general specify identical initial tapes. The source directories, original configurations, and PDF checksums are recorded in `figure/zoo-manifest.json`.

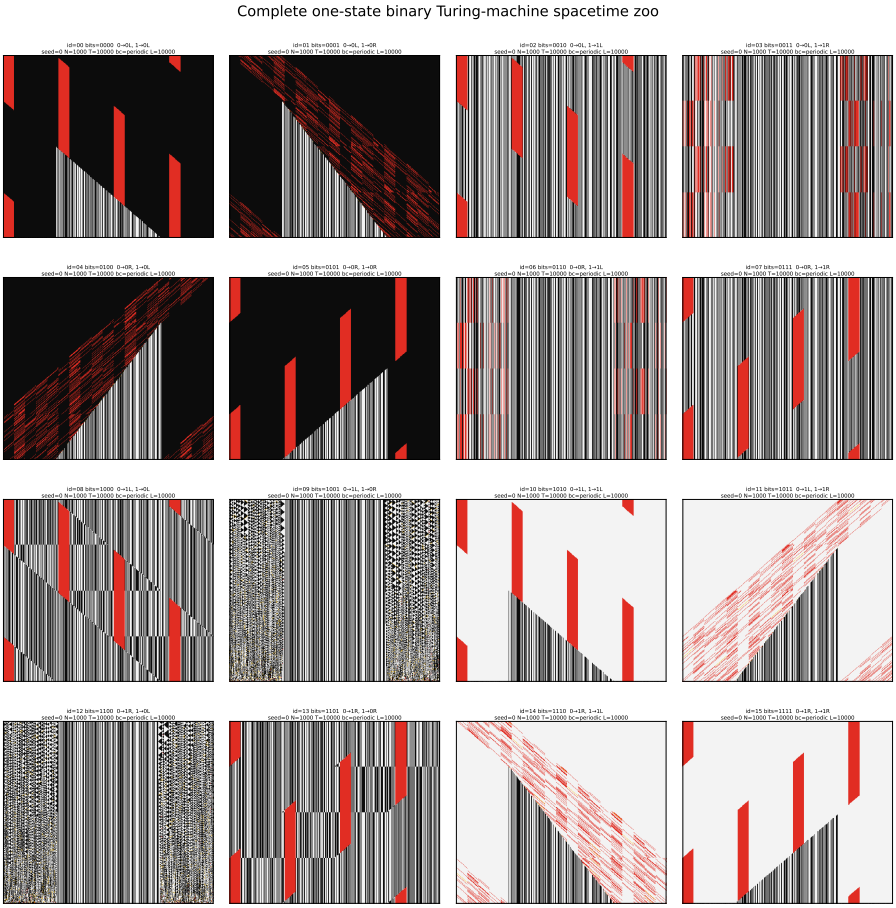


Figure C.1: Complete $\mathrm{TM}_1(2)$ zoo, IDs 0–15 in a 4×4 grid. Each panel uses $N = 1000$ identical heads, $q_t^{(i)} = 1$, $T = 10000$, $L = 10000$ with periodic boundaries, initial spacing five, and the same Bernoulli-(1/2) tape (seed 0). Updates are synchronous; all write proposals at a site agree. With $d_b = [M_1(b) + 1]/2$, the ID is the binary word $[W_1(0), d_0, W_1(1), d_1]$, and $S_1(b) = 1$. IDs $0, 2, 8, 10$ move only left and $5, 7, 13, 15$ only right; $3, 6$ preserve the tape and show localized motion here. IDs $1, 4, 11, 14$ write a constant symbol and show drifting fronts. IDs $9, 12$ flip symbols and move in opposite directions for the two reads; ID 9 is the rule in Section 1.1. Time increases upward; dark/light shows $V = 0/1$, with red/yellow marking single/multiple heads.

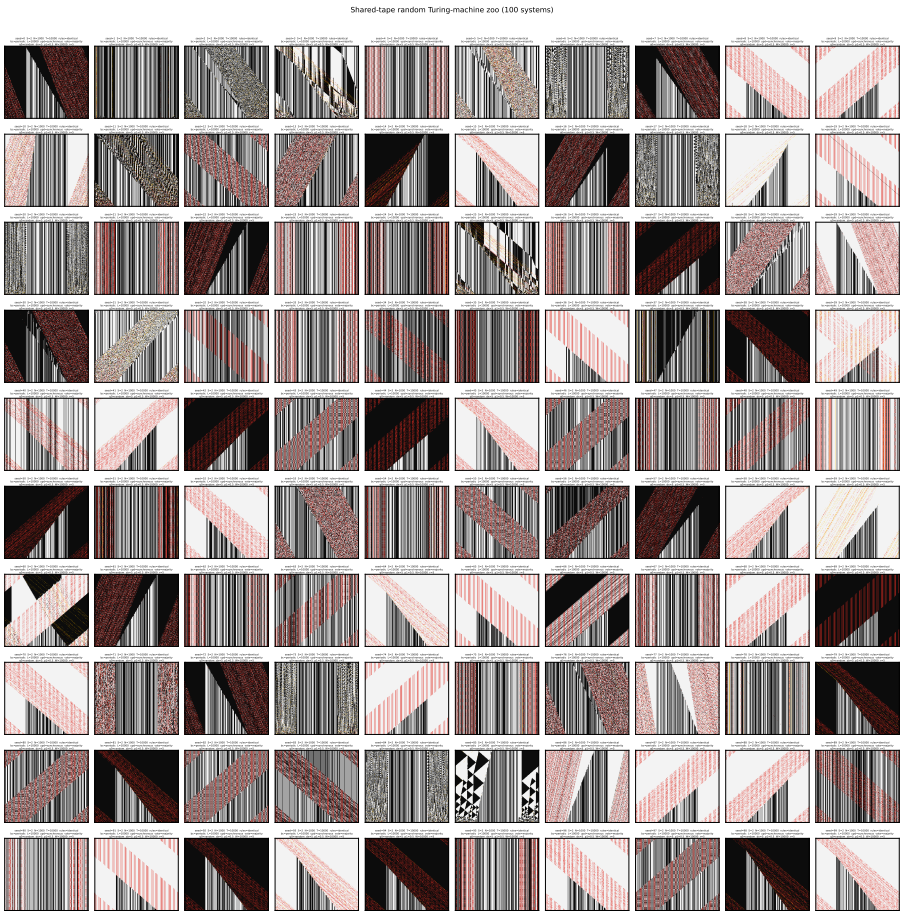


Figure C.2: Two-state machines with random initial head states. The 100 panels (seeds 0–99) each contain $N = 1000$ copies of one random $\mathrm{TM}_2(2)$, with $q_0^{(i)}$ uniform on $\{1, 2\}$ and a random tape with $p_1 = 1/2$. The ring has $L = 10000$, initial head spacing five, and $T = 10000$ synchronous steps with majority-vote writes and unchanged symbols on ties. The displayed width equals L, and snapshots were recorded every five steps. This samples two-state rules rather than enumerating them; repeated transition tables are possible. It extends Fig. C.1 to machines with internal memory. Time runs upward; tape values are dark/light and head occupancy is red/yellow.

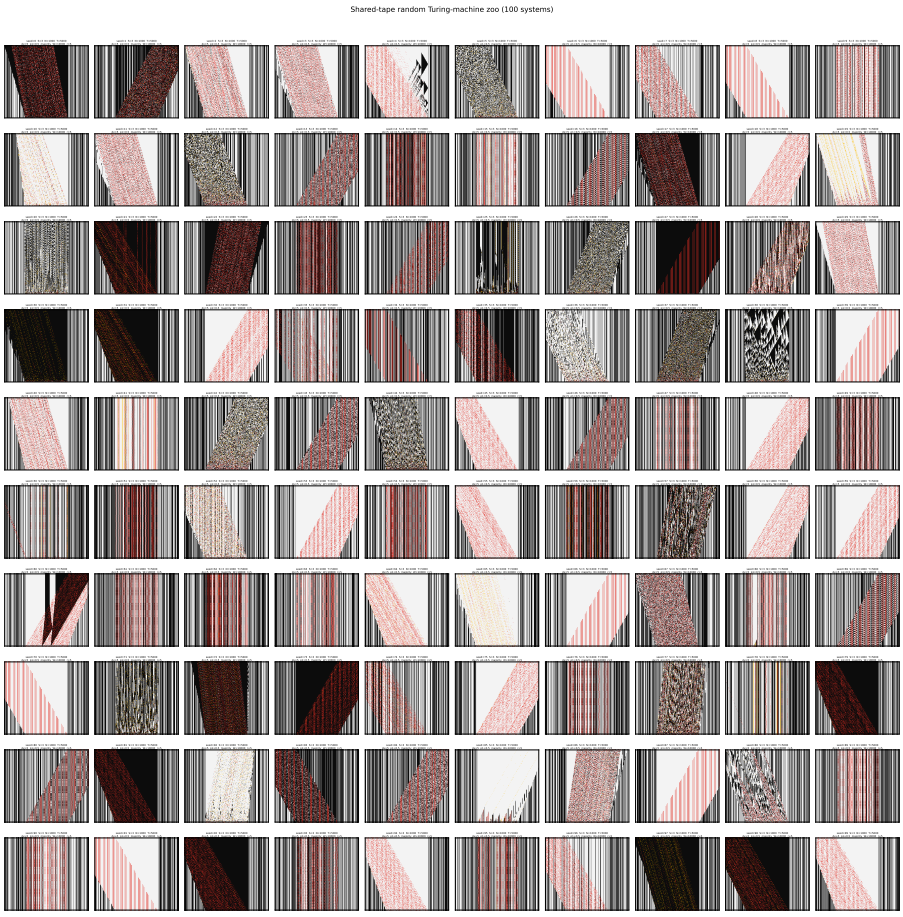


Figure C.3: Three-state reference ensemble: 100 random $\mathrm{TM}_3(2)$ tables, seeds 0–99, with $N = 1000$ identical copies per panel and $q_0^{(i)} = 1$. Each tape starts with independent bits of density $p_1 = 1/2$; heads are spaced five sites apart. Evolution lasts $T = 5000$ synchronous steps, with majority-vote writes and unchanged symbols on ties. The tape is unbounded and the displayed sites are $-5000 \le x \le 4999$; leaving the image does not remove a head. Snapshots were recorded every five steps. This is the reference for the additional seeds, blank tape, and asynchronous comparisons below. Time runs upward; dark/light denotes $V = 0/1$, and red/yellow denotes single/multiple heads.

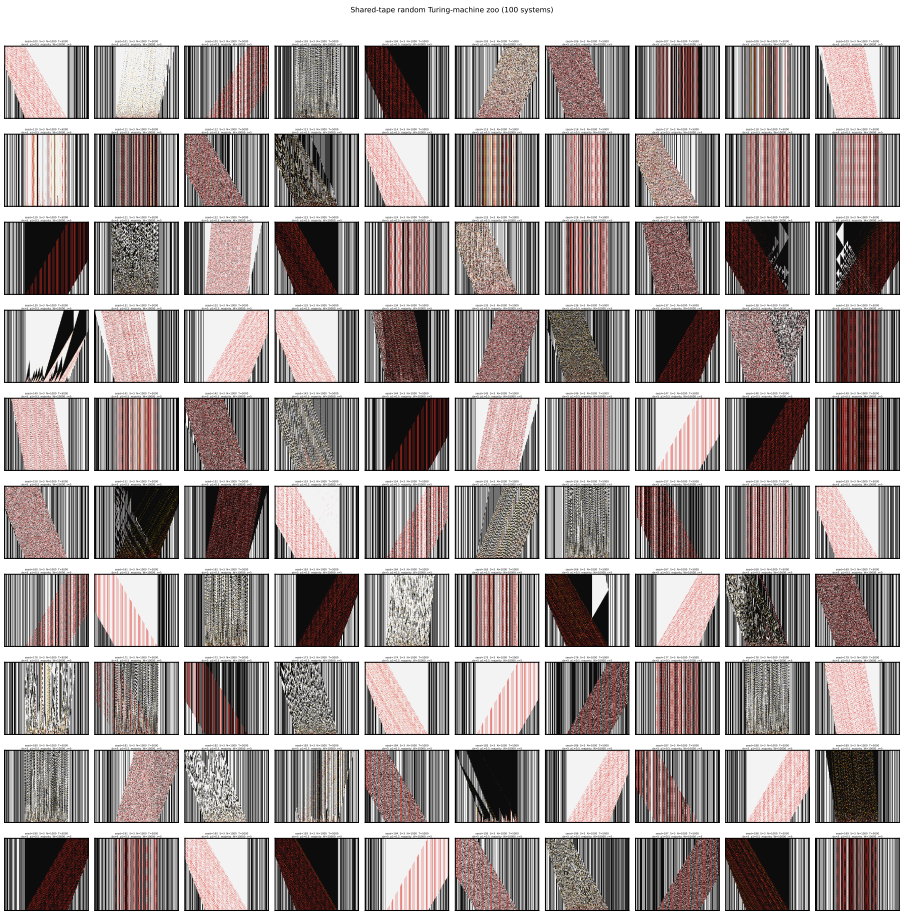


Figure C.4: A second sample of the three-state reference ensemble, using seeds 100–199 in reading order. Each of the 100 panels contains $N = 1000$ identical copies of a random $\mathrm{TM}_3(2)$, initialized at $q_0^{(i)} = 1$ on a Bernoulli-$(1/2)$ tape with head spacing five. The $T = 5000$ updates are synchronous, with majority-vote writes and unchanged symbols on ties; the unbounded tape is displayed over $-5000 \le x \le 4999$, with snapshots recorded every five steps. All ensemble parameters match Fig. C.3; the new seeds broaden the sample of tables and initial tapes, rather than continuing the earlier trajectories. Time runs upward, with the same tape and occupancy colors.

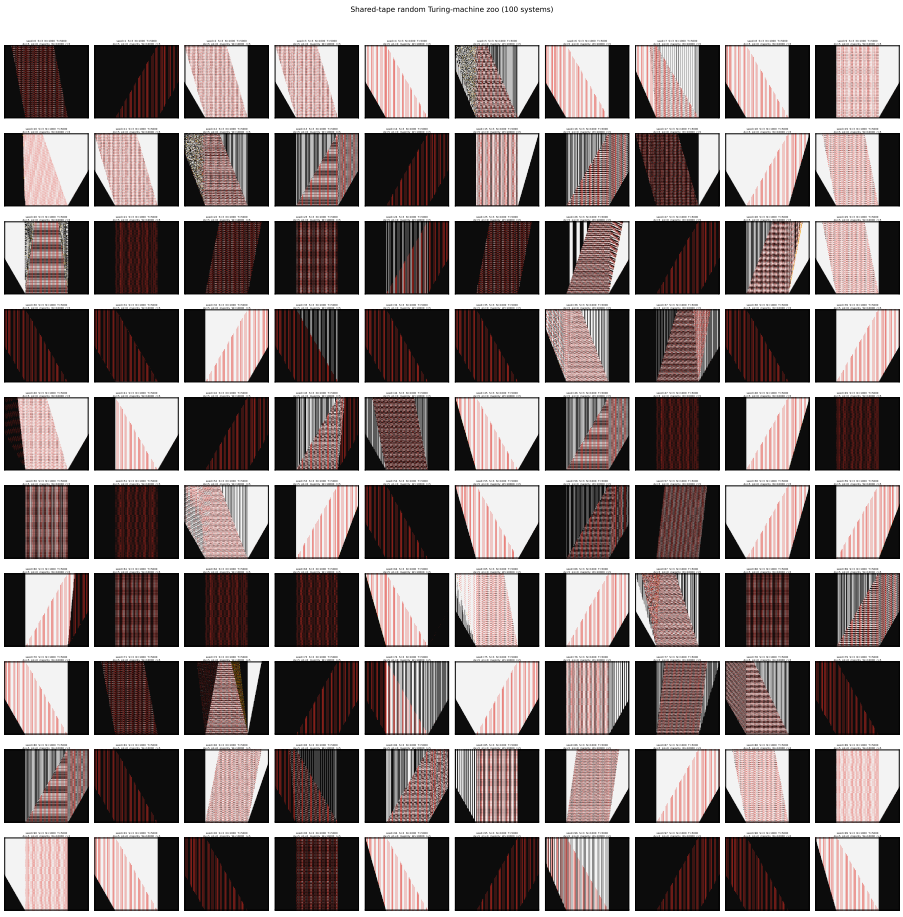


Figure C.5: Three-state machines on a blank tape, $V_0(x) = 0$. Seeds 0–99 use exactly the same $\mathrm{TM}_3(2)$ tables as Fig. C.3, with $N = 1000$ identical heads per panel, $q_0^{(i)} = 1$, and initial spacing five. The runs have $T = 5000$ synchronous steps, majority-vote writes with unchanged symbols on ties, and an unbounded tape displayed over $-5000 \le x \le 4999$. Snapshots were recorded every five steps. This comparison changes the initial landscape while retaining the transition tables; the blank background also makes regions modified by passing heads conspicuous. Time increases upward. Dark/light shows $V = 0/1$, and red/yellow shows single/multiple heads, not internal states.

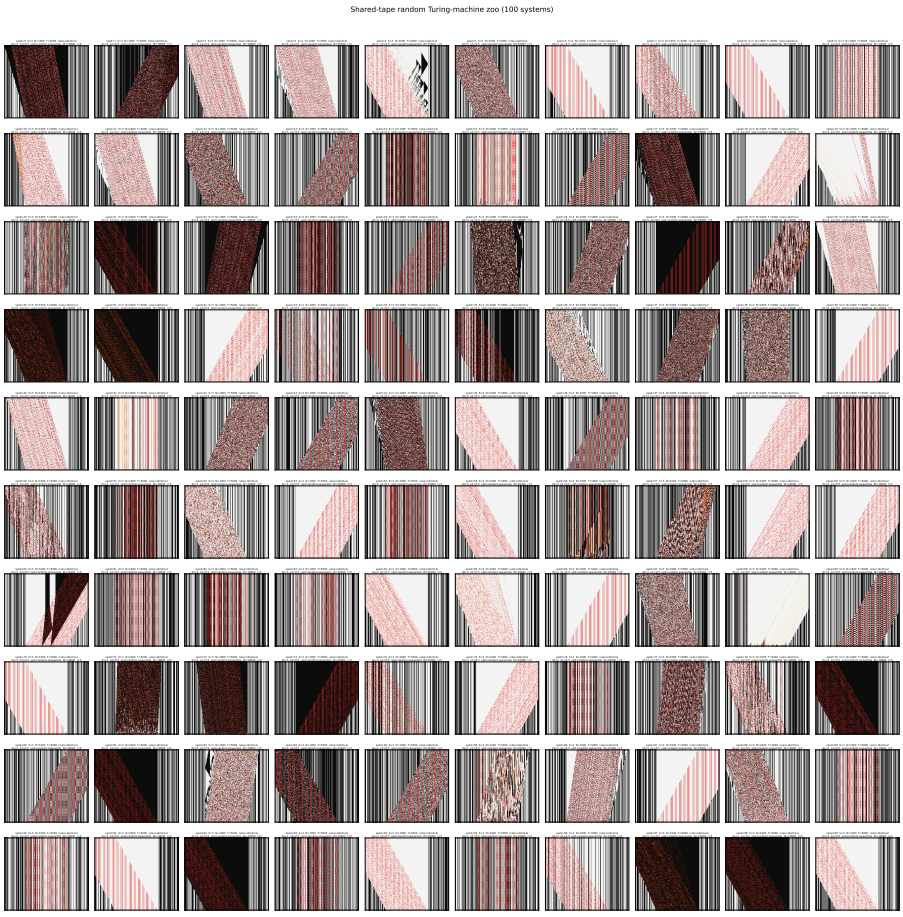


Figure C.6: Random-sequential evolution of identical $\mathrm{TM}_3(2)$ agents. The 100 panels, seeds 0–99, use the same transition tables as Fig. C.3, with $N = 1000$, $q_0^{(i)} = 1$, $p_1 = 1/2$, and initial spacing five. Here $T = 5000$ counts sweeps, each updating every head once in a fresh random order; writes take effect immediately, so majority voting is not applied. The tape is unbounded, the display covers $-5000 \le x \le 4999$, and snapshots were recorded every five sweeps. The comparison tests sensitivity to update order while preserving the individual machine rules. Time increases upward; the red/yellow overlay records occupancy at sweep boundaries and does not display the intermediate actions within a sweep.

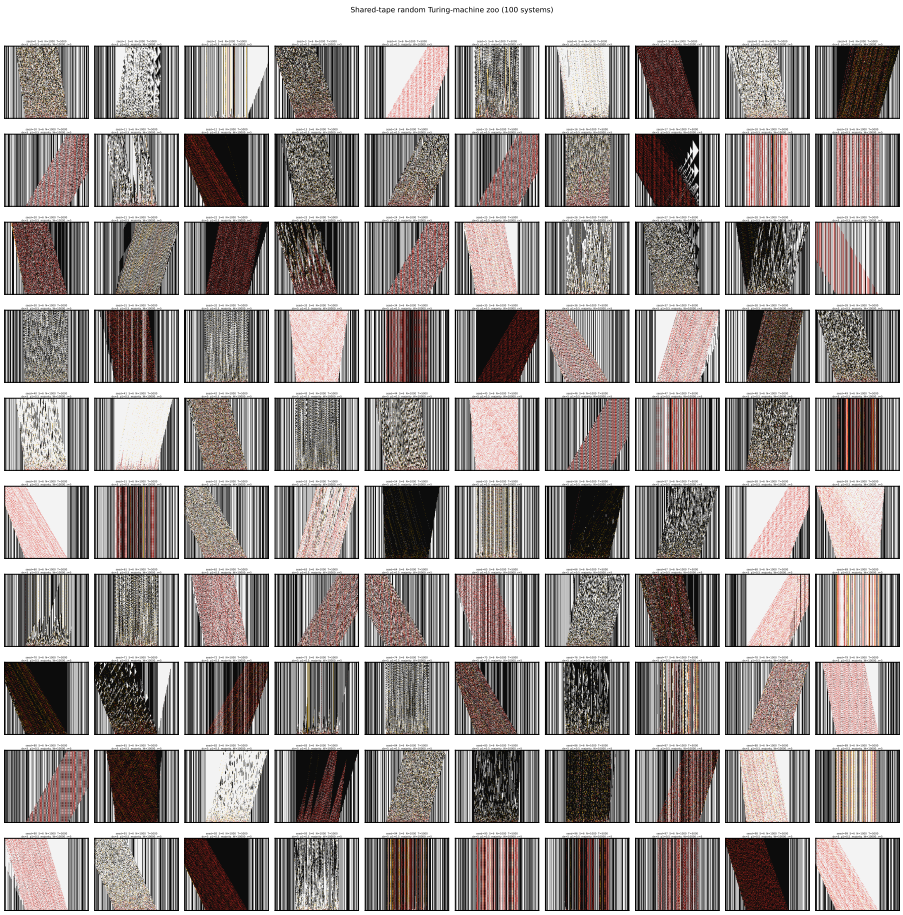


Figure C.7: Six-state reference ensemble. Each of the 100 panels, seeds 0–99, uses $N = 1000$ identical copies of a random $\mathrm{TM}_6(2)$, all initialized at $q_0^{(i)} = 1$. The initial tape is Bernoulli-(1/2) and the head spacing is five. Runs last $T = 5000$ synchronous steps with majority-vote writes and unchanged symbols on ties; the unbounded tape is shown over $-5000 \le x \le 4999$. Snapshots were recorded every five steps. Relative to the three-state ensemble, the state-space size changes and new transition tables are sampled. This grid is also the fixed-state reference for Fig. C.8. Time runs upward; dark/light indicates tape symbols and red/yellow indicates head occupancy.

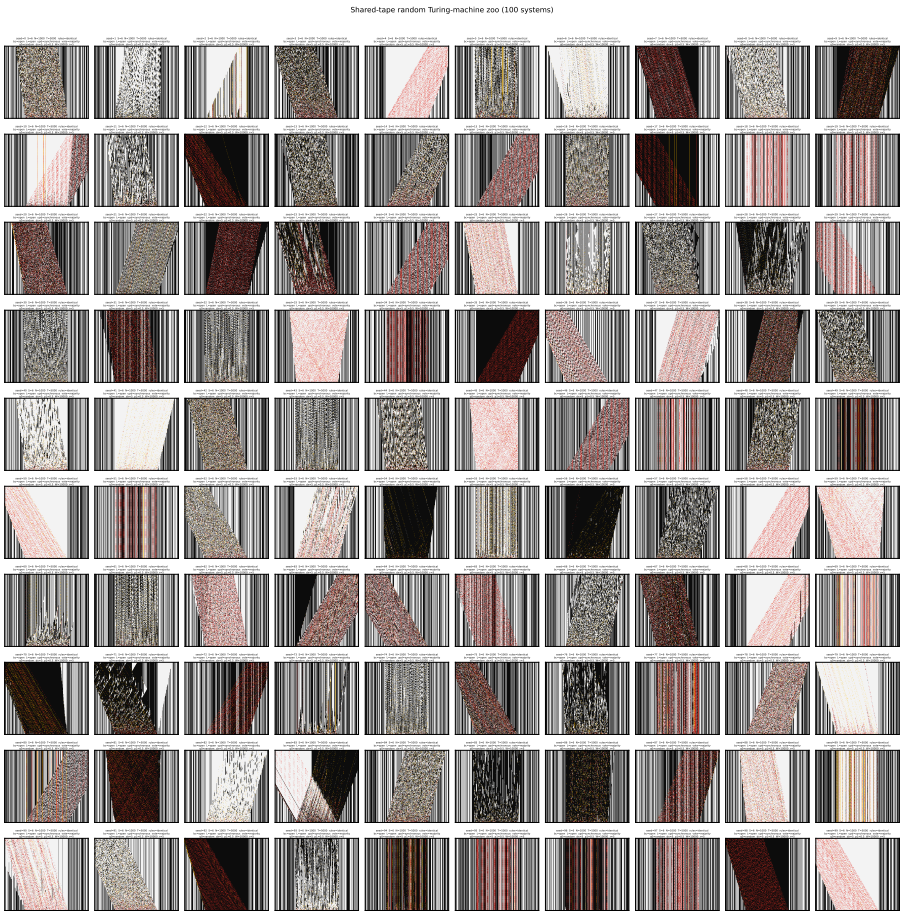


Figure C.8: Six-state machines with heterogeneous initial internal states. The 100 panels, seeds 0–99, share the same per-seed $\mathrm{TM}_6(2)$ tables as Fig. C.7, but now $q_0^{(i)}$ is independently uniform on [6]. Each run contains $N = 1000$ identical-rule heads, a Bernoulli-(1/2) tape, initial spacing five, and $T = 5000$ synchronous majority-vote steps (ties preserve the old symbol). The tape is unbounded; the plotted sites are $-5000 \le x \le 4999$, recorded every five steps. This separates diversity of initial head states from diversity of transition tables. Time runs upward; the red/yellow overlay indicates single/multiple heads and does not encode their internal states.

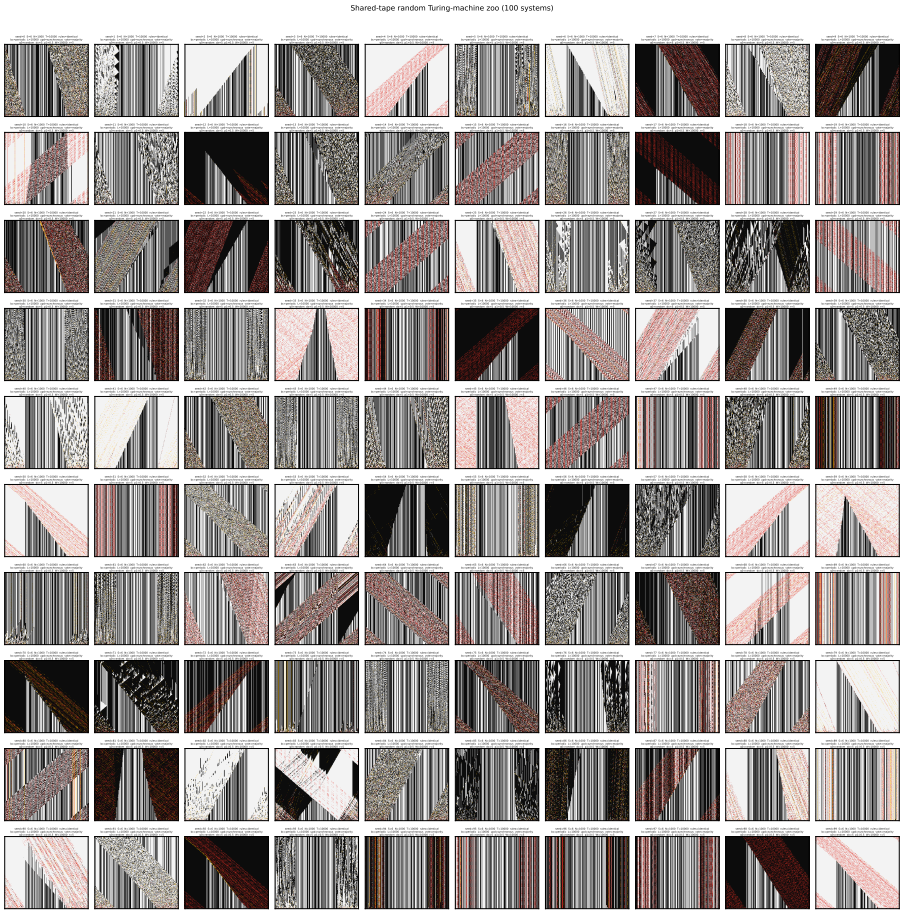


Figure C.9: Longer periodic runs of the six-state, random-initial-state ensemble. Seeds 0–99 retain the transition tables of Fig. C.8, with $N = 1000$ identical $\mathrm{TM}_6(2)$ heads, $q_0^{(i)}$ uniform on $[6]$, $p_1 = 1/2$, and initial spacing five. Here the full ring $L = 10000$ is shown for $T = 10000$ synchronous steps, using majority-vote writes with unchanged symbols on ties and recording every five steps. Both the geometry and duration differ from the unbounded $T = 5000$ runs; these are new periodic realizations, not continuations of the earlier tapes. Time runs upward. Trajectories can wrap across the spatial edges; those edges are not absorbing boundaries.

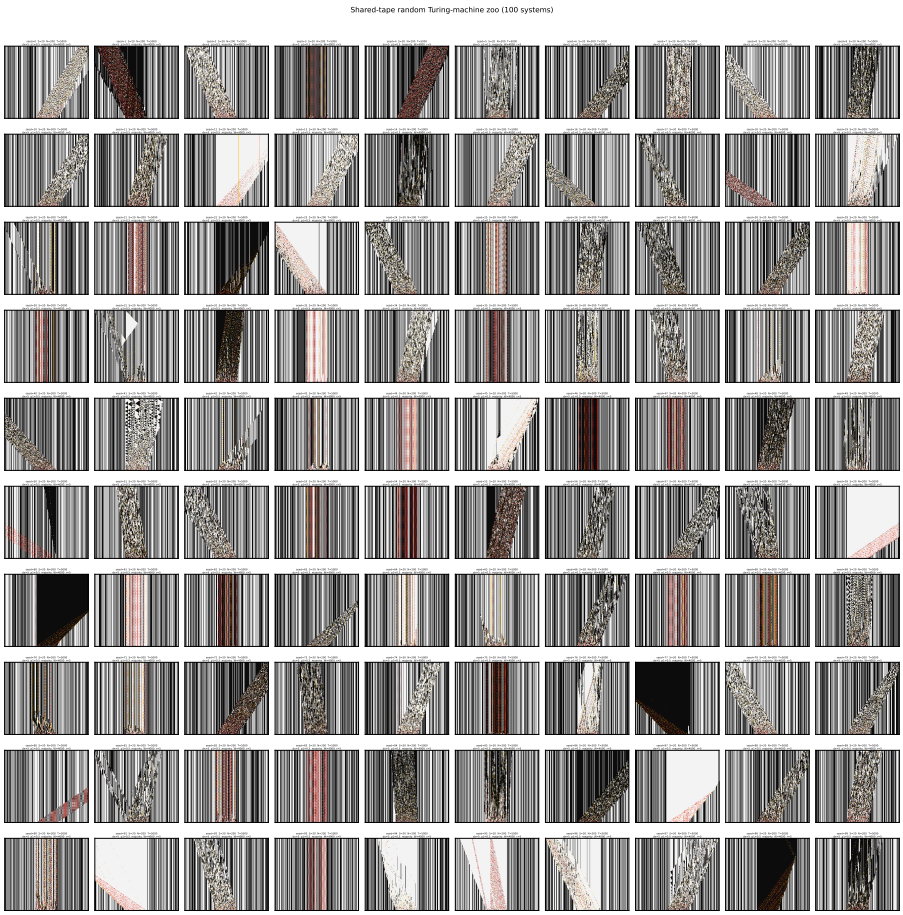


Figure C.10: Original twenty-state zoo with $N = 200$ heads. Each of the 100 panels, seeds 0–99, uses one random $\mathrm{TM}_{20}(2)$ shared by all heads, with $q_0^{(i)} = 1$, initial spacing five, and an independent-bit tape with $p_1 = 1/2$. Evolution lasts $T = 5000$ synchronous steps with majority-vote writes and unchanged symbols on ties. The tape is unbounded and the narrower display contains $-2000 \le x \le 1999$; snapshots were recorded every five steps. This is the smaller-population reference for Fig. C.11. Time runs upward; dark/light shows tape symbols and red/yellow shows single/multiple heads. Heads crossing the display edge continue to evolve outside the image.

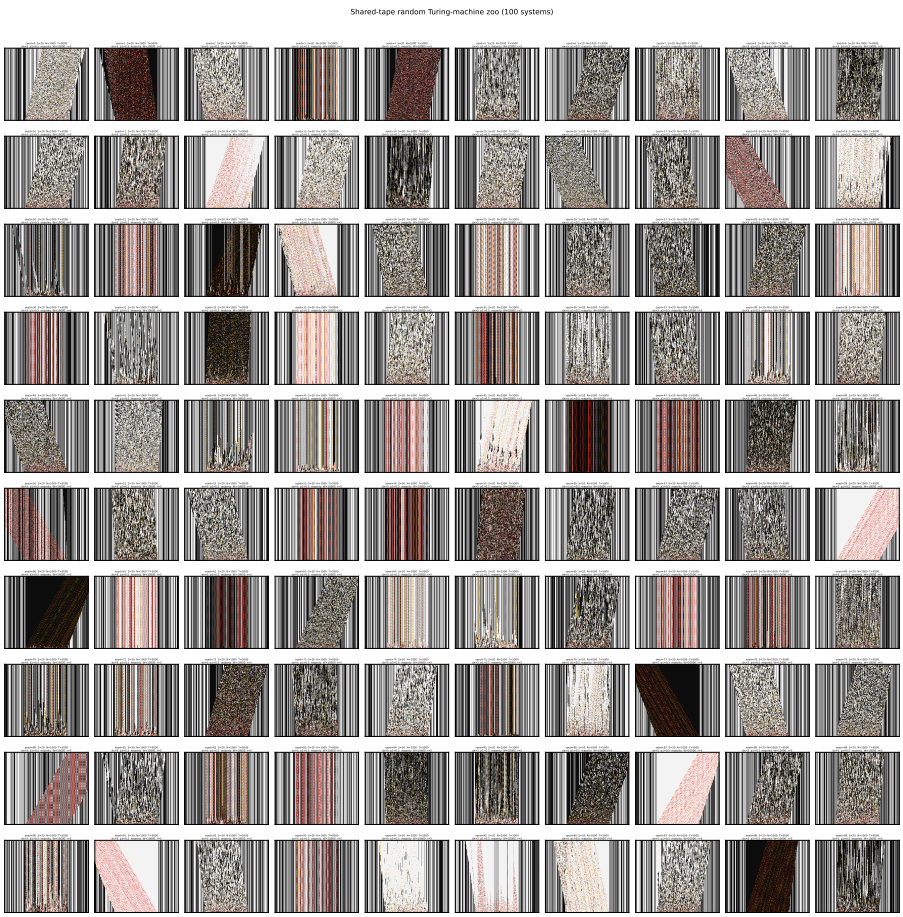


Figure C.11: Twenty-state zoo with the population increased to $N = 1000$. The 100 panels, seeds 0–99, contain identical-rule $\mathrm{TM}_{20}(2)$ heads with $q_0^{(i)} = 1$, initial spacing five, and $p_1 = 1/2$. The $T = 5000$ updates are synchronous, using majority-vote writes and unchanged symbols on ties. The unbounded tape is now shown over $-5000 \le x \le 4999$, with snapshots recorded every five steps. Compared with Fig. C.10, the initial occupied interval and display window are larger while the local head spacing is unchanged; this is not an increase of N/L on a fixed periodic ring. Time runs upward, with dark/light tape values and red/yellow occupancy.

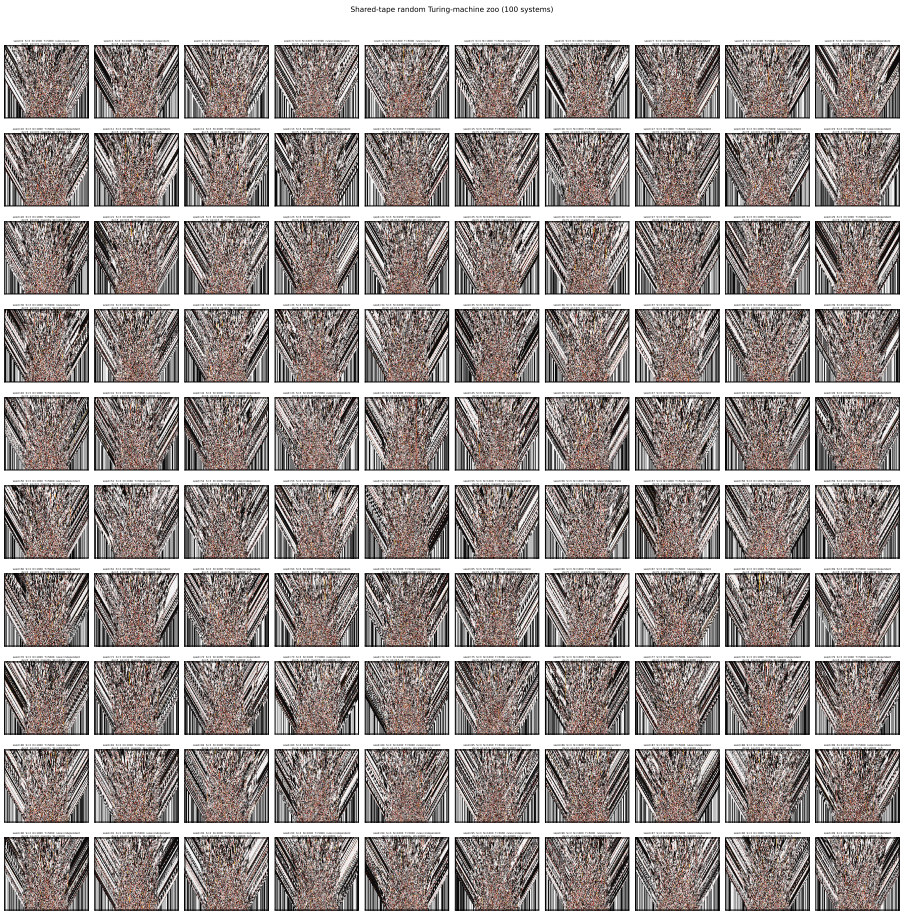


Figure C.12: A heterogeneous gas of three-state machines on an unbounded tape. For each seed 0–99, the $N = 1000$ heads receive independently sampled $\mathrm{TM}_3(2)$ transition tables $(M^{(i)}, W^{(i)}, S^{(i)})$; all have $q_0^{(i)} = 1$. Initial spacing is five and $p_1 = 1/2$. Each run lasts $T = 5000$ synchronous steps with majority-vote writes and unchanged symbols on ties. The displayed window is $-5000 \le x \le 4999$, with snapshots recorded every five steps. Unlike Fig. C.3, disorder now occurs between heads within each realization as well as between panels. The tables stay fixed in time. Time runs upward; the common red/yellow overlay marks occupancy and does not distinguish the individual machine rules.

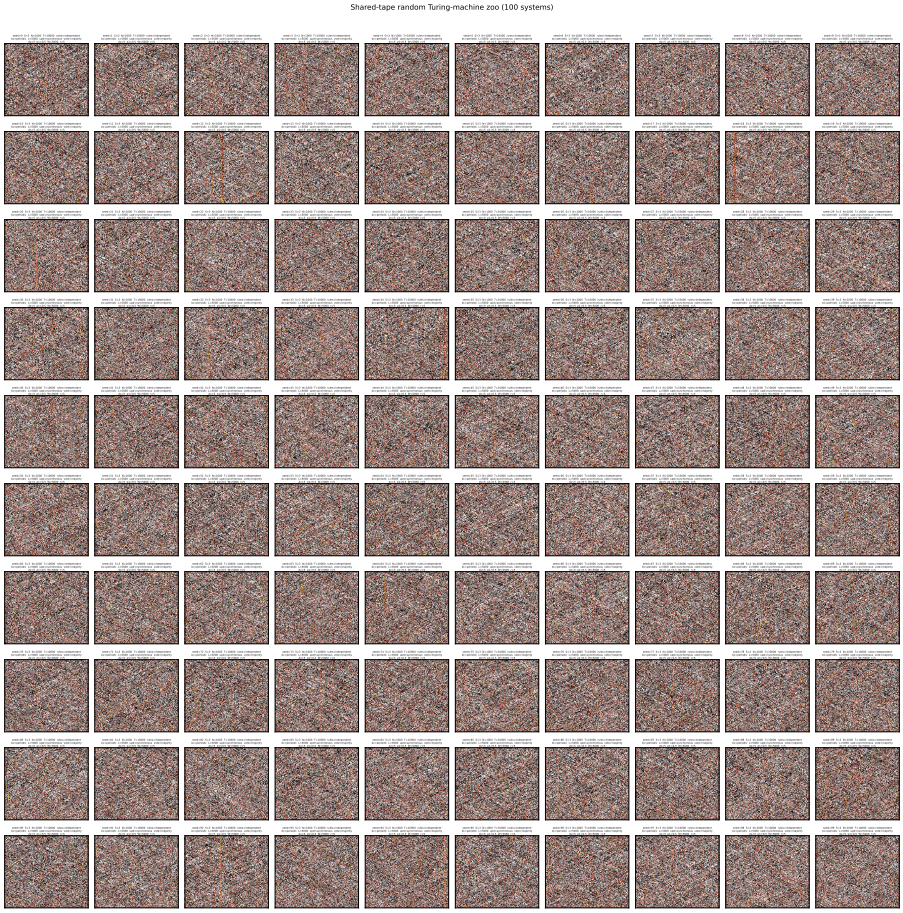


Figure C.13: Heterogeneous three-state gas on a shorter periodic tape. Each seed 0–99 samples $N = 1000$ independent $\mathrm{TM}_3(2)$ tables, with $q_0^{(i)} = 1$, $p_1 = 1/2$, and initial spacing five. The full ring has $L = 5000$, hence $N/L = 0.2$, and the duration is $T = 10000$ synchronous steps. Writes use majority vote with unchanged symbols on ties; snapshots were recorded every five steps. Relative to Fig. C.12, the duration doubles and the displayed spatial extent halves, while periodic wrapping replaces the unbounded geometry. The texture shown by these overviews alone does not establish thermalization. Time runs upward; dark/light represents tape symbols and red/yellow represents single/multiple heads.

D Halting-time distributions: exact identities and a route to a solution

This section concerns a single machine on its own tape, rather than the interacting gas. The aim is to derive the distribution of halting times over random transition tables and initial tapes. We distinguish exact finite-time results from a proposed program for the long-time tail; an approximately straight histogram on logarithmic axes is not by itself a proof of either an exponential or a power-law distribution.

D.1 Ensemble and halting-time convention

Use the functions $W_q(b)$, $M_q(b)$, and $S_q(b)$ of Section 1, with n working states and two symbols. The head starts at $x_0 = 0$ in state $q_0 = 1$ on the unbounded tape $x \in \mathbb{Z}$. Initially either $V_0(x) = 0$ everywhere or the symbols are independent Bernoulli-(1/2) variables. The latter randomness is sampled once: revisiting a cell reads its current value, not a fresh coin. Introduce a halting state $H \notin [n]$ and define

$$T = \inf\{t \geq 1 : q_t = H\}, \qquad f_n(t) = \Pr(T = t), \qquad F_n(t) = \Pr(T \leq t). \quad (67)$$

Set $T = \infty$ if the machine never halts. One step comprises reading, writing, moving, and changing state, including the operation entering H; thus the first possible halting time is $T = 1$.

Our sampling convention is uniform over labeled tables with exactly one halting entry. Equivalently, first sample a non-halting parent table with independent uniform choices of $W_q(b) \in \{0, 1\}$, $M_q(b) \in \{-1, +1\}$, and $S_q(b) \in [n]$. Independently mark one entry $J \in [n] \times \{0, 1\}$ uniformly, and replace only S_J by H. There are $8n(4n)^{2n-1}$ resulting labeled tables: $2n$ choices of halting entry, four write/move choices there, and $4n$ choices at each other entry. Unreachable states are allowed, and no symmetry quotient or connectivity conditioning is imposed. The formulas below refer to this measure, not to independently assigning a halt target to every entry.

In general $\sum_{t \geq 1} f_n(t) < 1$, with the remaining mass at infinity. A histogram normalized over runs that halt before a cutoff $t_{\max}$ estimates $f_n(t)/F_n(t_{\max})$ for $t \leq t_{\max}$, not necessarily the distribution conditional on eventual halting. In particular, survival to a cutoff on a random tape is not a non-halting certificate.

D.2 An exact reduction to discovery of transition entries

Run the non-halting parent on the chosen initial tape. Write $\widetilde{q}_j$, $\widetilde{x}_j$, and $\widetilde{V}_j$ for its trajectory, and define

$$\mathcal{E}_t = \big\{(\widetilde{q}_j, \widetilde{V}_j(\widetilde{x}_j)) : 0 \leq j < t\big\}, \qquad K_t = |\mathcal{E}_t|, \qquad K_0 = 0. \quad (68)$$

Thus K_t counts distinct state/read-bit entries used in the first t operations, not distinct internal states or tape positions.

Exact identity and proof. For a fixed parent table and initial tape, the modified and parent trajectories agree until the first use of J. At that operation the modified machine halts. Therefore $T \le t$ if and only if $J \in \mathcal{E}_t$. Averaging the independent uniform mark, then the parent and tape, gives

$$\boxed{F_n(t) = \frac{\mathbb{E}[K_t]}{2n},} \qquad \boxed{f_n(t) = \frac{\mathbb{E}[K_t - K_{t-1}]}{2n}.} \tag{69}$$

The same proof works for either initial-tape ensemble. Expectations here are over non-halting parents and their initial tapes; the parent is continued indefinitely even after a marked version would have halted.

Since K_t increases to a bounded random variable $K_\infty \le 2n$,

$$\Pr(T < \infty) = \frac{\mathbb{E}[K_\infty]}{2n}, \tag{70}$$

$$\boxed{\Pr(t < T < \infty) = \frac{\mathbb{E}[K_\infty - K_t]}{2n}.} \tag{71}$$

The finite halting tail is consequently the expected number of entries that will eventually be discovered but have not yet been discovered. It is not proportional to the number $2n - K_t$ of all unseen entries, some of which may remain inaccessible forever. This also explains why an independent, constant halting hazard is not automatic: a previously used non-halting entry stays safe on every subsequent use.

D.3 First exact coefficients and numerical checks

The first operation always discovers an entry, so for either tape ensemble

$$f_n(1) = \frac{1}{2n}. \tag{72}$$

For the random tape, the second operation reads an independent fair bit at a new site. The parent's first next state is independently uniform on $[n]$. Thus the second state/read pair equals the first with probability $1/(2n)$, and Eq. (69) gives

$$f_n(2) = \frac{1}{2n}\left(1 - \frac{1}{2n}\right) = \frac{2n-1}{4n^2} \qquad \text{(iid fair random tape)}. \tag{73}$$

On a blank tape both of these reads are zero, so the second pair repeats the first with probability $1/n$. Hence

$$f_n(2) = \frac{n-1}{2n^2} \qquad \text{(blank tape)}. \tag{74}$$

These expressions are unconditional probabilities, including all sampled machines in the denominator. For $n = 4$ on a random tape they predict $f_4(1) = 0.125$

and $f_4(2) = 0.109375$. The existing 10^6-sample run (machine seed 0, tape seed 1, cutoff 10000) gives 0.124794 and 0.109275. The counts and configuration are recorded in `results/halt_times_4state_2symbol_random_tape/distribution.csv` and the adjacent `summary.json`. This agreement is a numerical check of the derivation, not a proof of the later coefficients.

D.4 Exact enumeration by deferred sampling

A constructive extension is to explore weighted execution histories, revealing each random choice only when it is first needed. This uses the principle of lazy sampling [16] and is related to probabilistic symbolic execution, which computes probabilities of execution paths [17]. These are methodological connections; those references do not solve the present TM ensemble.

For direct exploration of the halting ensemble, retain the current head position and state, the current tape on visited sites, and the partially revealed rule table. At any surviving history with k distinct known non-halting entries, the unique halting entry is uniform among the $2n-k$ remaining entries. The exact branching rules are:

1. On first reading an unvisited random-tape cell, branch over its two initial symbols with weights 1/2. For a blank tape its initial symbol is deterministically zero. Visited cells retain their current values, including all previous writes.

2. If the state/read pair is already known, execute its stored rule deterministically; do not resample it or give it another chance to halt.

3. If the pair is new, halt with conditional probability $1/(2n-k)$. Otherwise branch over the $4n$ possible non-halting actions. Each such action has conditional probability
$$\frac{2n-k-1}{(2n-k)\,4n}, \tag{75}$$
and is stored permanently. The final write and move on a halting branch can be marginalized because they do not affect T.

Multiplying these probabilities along branches and summing the weight absorbed at operation t computes $f_n(t)$ exactly. An induction on execution length establishes equivalence to sampling the whole table and initial tape in advance. Every fixed horizon has finitely many branches and rational weights; the difficulty is their number, not an unresolved probability measure.

Equivalent explored configurations may be merged if their distributions of all unrevealed choices also agree. Translation and appropriate state relabelings can reduce duplication, provided branch multiplicities are preserved. In contrast, keeping only q_t, the read bit, or K_t does not generally close the dynamics: spatial tape memory determines which entry can be discovered next. The exact enumerator proposed here has not yet been implemented.

D.5 An exactly solvable right-moving subclass

Fix a table satisfying $M_q(b) = +1$ for every entry, with a fair iid initial tape. Every operation reads a fresh independent bit, and written symbols are never read again. The working state alone is therefore an absorbing Markov chain. With columns denoting the initial state, define

$$Q_{ij} = \frac{1}{2}\sum_{b=0}^{1} \mathbf{1}_{\{S_j(b)=i\}}, \qquad h_j = \frac{1}{2}\sum_{b=0}^{1} \mathbf{1}_{\{S_j(b)=H\}} = 1 - \sum_{i=1}^{n} Q_{ij}. \tag{76}$$

Here Q is substochastic and e_1 denotes unit mass in state 1. For this fixed table the tape-averaged distribution and its generating function are exactly

$$\Pr(T = t \mid \text{table}) = h^{\mathsf{T}} Q^{t-1} e_1, \tag{77}$$

$$G(z) = \sum_{t\geq 1} \Pr(T = t \mid \text{table}) z^t = z h^{\mathsf{T}} (I - zQ)^{-1} e_1, \qquad |z| < 1. \tag{78}$$

Closed non-halting classes can leave mass at infinity. Matrix powers can give several modes and periodicity, not a single geometric law. Averaging over right-moving tables solves that subclass, not the full ensemble, and supplies an exact benchmark for enumeration.

D.6 What remains to prove about the long-time tail

For general machines, tape revisits and overwriting invalidate the fresh-symbol assumption behind Eq. (78). An internal-state transition graph alone cannot repair this loss of memory. A plausible research program is:

1. Compute exact short-time coefficients and compare with Monte Carlo frequencies, keeping unconditional probabilities, cutoff conditioning, and certified non-halting mass distinct.
2. Examine long-lived parent trajectories for delayed first uses of entries. Identify recurring tape patterns, repeated excursions, or growing counters that can delay a new entry without making it permanently inaccessible.
3. For each class, prove tape-pattern recurrences or a renewal description at regeneration points. Retained variables must encode all relevant memory; return to the same internal state alone is insufficient.
4. Combine the class probabilities and their first-discovery laws. A tail theorem for the full ensemble must also bound any unclassified remainder, rather than extrapolate the dominant sampled behavior.

By Eq. (71), a proved bound $\mathbb{E}[K_\infty - K_t] \leq Ce^{-\lambda t}$ would imply $\Pr(t < T < \infty) \leq Ce^{-\lambda t}/(2n)$; fitting an exponential to a histogram does not establish this bound. The discovery identities, first coefficients, and right-moving solution are exact. Efficient enumeration and a general tail law remain proposed work, not established solutions.

E Mutual rewriting of two program tapes

In the gases above the transition table is a fixed parameter and only the shared tape evolves. Here the table itself becomes the dynamical variable. Two machines in $\mathrm{TM}_2(2)$ each store their own transition table on a short periodic tape, and each machine's head reads and writes the tape that encodes the *other* machine. The question is whether some tables are attracting or repelling under this mutual rewriting. The setting is closer to algorithmic chemistry [7] than to the shared-tape gas, since the objects acted upon are themselves programs.

Because the joint state space is finite, everything reported in this section is an exact enumeration over the stated initial conditions, not a sampled estimate. The mechanism offered for the dominant attractors is an interpretation and is marked as such.

E.1 Model and encoding

A table in $\mathrm{TM}_2(2)$ has four entries (q, b), each specifying a write bit, a move, and a next state, hence $3 \times 4 = 12$ bits and $2^{12} = 4096$ labeled tables. Machine $a \in \{1, 2\}$ carries a periodic tape $\tau_t^{(a)} \in \{0, 1\}^{12}$ with cells indexed by $\mathbb{Z}_{12}$. Entry (q, b) has index $k = 2(q-1) + b$ and occupies three consecutive cells,

$$\begin{aligned} W_t^{(a)}(q, b) &= \tau_t^{(a)}(3k), \\ M_t^{(a)}(q, b) &= 2\,\tau_t^{(a)}(3k+1) - 1, \\ S_t^{(a)}(q, b) &= 1 + \tau_t^{(a)}(3k+2), \end{aligned} \tag{79}$$

so that $(W_t^{(a)}, M_t^{(a)}, S_t^{(a)})$ are the functions $W_q(b)$, $M_q(b)$, $S_q(b)$ of Section 1 for the table currently written on tape a. A stored move bit 0 means left and 1 means right; a stored state bit 0 means state 1.

Write $\bar{a}$ for the other machine. The head of machine a sits at cell $x_t^{(a)} \in \mathbb{Z}_{12}$ of tape $\bar{a}$ in state $q_t^{(a)} \in \{1, 2\}$. With $b_t^{(a)} = \tau_t^{(\bar{a})}(x_t^{(a)})$ the symbol it reads, one step is

$$\tau_{t+1}^{(\bar{a})}\big(x_t^{(a)}\big) = W_t^{(a)}\big(q_t^{(a)}, b_t^{(a)}\big), \tag{80}$$

$$x_{t+1}^{(a)} = x_t^{(a)} + M_t^{(a)}\big(q_t^{(a)}, b_t^{(a)}\big) \pmod{12}, \tag{81}$$

$$q_{t+1}^{(a)} = S_t^{(a)}\big(q_t^{(a)}, b_t^{(a)}\big), \tag{82}$$

for $a = 1, 2$ simultaneously, all other cells being unchanged. Both heads read the tapes at time t and both use the tables decoded from the tapes at time t; the two writes then land together. The heads write on different tapes, so no collision rule is needed. A write can alter the very entry that the other machine uses at the same step; that machine still acts by the old entry and the new one takes effect at $t + 1$.

The joint state consists of the two tapes, the two head cells, and the two head states. It takes

$$4096^2 \cdot 12^2 \cdot 2^2 = 9{,}663{,}676{,}416 \tag{83}$$

values and evolves deterministically, so every initial condition reaches a cycle after a finite transient. The reference ensemble consists of all $4096^2 = 16{,}777{,}216$ ordered pairs of tables, with both heads started at cell 0 in state 1. Exchanging the labels of the two machines is an exact symmetry of the dynamics.

E.2 Attractors and per-table observables

For each initial pair we compute the exact transient length and period of the joint state. An *attractor* is a cycle of the joint state. A program is *frozen* on an attractor if its tape is the same at every step of the cycle; the heads continue to move, so a frozen pair still has a nontrivial period set by head motion. Initial pairs are sorted into five outcomes: both programs frozen at the all-0 tape; both frozen at the all-1 tape; both frozen at the tables they started with; any other frozen pair; and cycles on which at least one program keeps changing. The third class is defined by comparing the final and initial tables, and does not distinguish a program that was never written to from one that was altered and restored during the transient.

Three quantities are attached to each table T. Let $\mathcal{I}$ be the set of initial pairs and P_i the period reached from $i \in \mathcal{I}$. The *late-time occupancy* is

$$\omega(T) = \frac{1}{2|\mathcal{I}|} \sum_{i \in \mathcal{I}} \frac{1}{P_i} \sum_{t \in \mathrm{cycle}(i)} \Big[\mathbf{1}_{\{\tau_t^{(1)} = T\}} + \mathbf{1}_{\{\tau_t^{(2)} = T\}} \Big], \qquad \sum_T \omega(T) = 1, \quad (84)$$

so a table that is neither favored nor disfavored has $\omega = 1/4096$. The *survival fraction* $\sigma(T)$ is the fraction of the 8192 combinations of initial pair and machine label in which T is present initially and that machine ends frozen at T. The *capture count* $c(T)$ is the number of such combinations in which T was not present initially and the machine nevertheless ends frozen at T. We call a table attracting when $\omega(T) \gg 1/4096$ with large $c(T)$, and repelling when it has $\omega(T) = 0$ or $\sigma(T) = 0$. These notions are relative to the uniform measure on initial pairs.

E.3 Exact structural statements

Two families of fixed pairs follow directly from Eq. (79).

Uniform tapes. The all-0 tape decodes to $W \equiv 0$, $M \equiv -1$, $S \equiv 1$: write 0, move left, enter state 1. Acting on an all-0 tape it changes nothing, so the pair (all-0, all-0) is frozen with period 12. The all-1 tape decodes to $W \equiv 1$, $M \equiv +1$, $S \equiv 2$, and (all-1, all-1) is likewise frozen. Each of these programs writes the symbol of which its own code consists.

Tables that write what they read. There are $4^4 = 256$ tables with $W_q(b) = b$ for all four entries. Such a machine never changes the tape it acts on. If both machines are of this kind then neither tape ever changes, so all $256^2 = 65{,}536$ such pairs are frozen at their initial tables from $t = 0$.

E.4 Results for the reference ensemble

Table 1 gives the outcome of every initial pair. Ninety percent of pairs end with both programs equal to one of the two uniform tapes, and 90.11% end with two identical programs. The transient has mean 35.6 steps, median 30, and maximum 397. The pairs reach 791,425 distinct cycles: 679,535 frozen cycles, which cover 539,731 distinct pairs of programs and otherwise differ in head phase, and 111,890 cycles on which a program keeps changing. The longest period is 132, attained by a frozen pair.

Table 1: Outcomes of all 16,777,216 ordered pairs of two-state tables under mutual rewriting, with both heads started at cell 0 in state 1.

Outcome	Initial pairs	Share
Both programs frozen at the all-0 tape	8,266,669	49.27%
Both programs frozen at the all-1 tape	6,832,632	40.73%
Both programs frozen at their initial tables	407,333	2.43%
Other frozen pair	912,052	5.44%
At least one program keeps changing	358,530	2.14%

Attracting tables. The all-0 tape has $\omega = 0.4927$, about 2018 times the uniform share, and the all-1 tape has $\omega = 0.4074$, about 1669 times. They capture 16,527,452 and 13,659,600 of the $2 \times 16{,}777{,}216$ combinations of pair and machine. Neither is invulnerable: $\sigma = 71.9\%$ for the all-0 tape and 69.1% for the all-1 tape, so in more than a quarter of the pairs containing one of them it does not survive. For example, the initial pair (all-0, all-1) ends as (all-1, all-1) after 15 steps. The only other table with a notable occupancy is the alternating tape 010 101 010 101, at 12.3 times the uniform share with $\sigma = 18.0\%$ and $c = 94{,}720$; it is one of the tables that write what they read. No fourth table exceeds 3.5 times the uniform share.

Repelling tables. There are 162 tables with $\omega = 0$, which are never present on any cycle reached from the reference ensemble. A much larger set never persists: 2836 of the 4096 tables have $\sigma = 0$, and 2792 have $c = 0$.

Classes by write bits. Table 2 groups the tables by their four write bits. The tables that write what they read are robust without being attracting: their mean survival is several times that of any other class, none of them has $\omega = 0$, and together they hold only 1.9% of the late-time occupancy. The constant-writing classes hold almost all of the occupancy, but nearly all of it sits on the two uniform tapes; the mean survival within those classes is low, so a constant writer other than a uniform tape rarely ends frozen at its own table.

Table 2: Tables grouped by their write bits. Occupancy is the summed $\omega(T)$ of the class; survival is the class mean of $\sigma(T)$.

Class	Tables	Occupancy	Mean survival	Tables with $\omega = 0$
$W \equiv 0$	256	49.65%	2.0%	7
$W \equiv 1$	256	41.10%	1.0%	3
$W_q(b) = b$	256	1.93%	15.4%	0
All others	3328	7.31%	2.9%	152

The outcome as a vote on write bits. Table 3 conditions the outcome on the number of write-1 entries in the two initial tables together. The probability of ending at the all-1 pair increases monotonically with that count, and at an even split the all-0 pair is favored by ten percentage points. The extreme rows are not exactly 100%: two machines that only write 0 can freeze with stored 1s in cells that neither head visits again.

Table 3: Outcome conditioned on the number of write-1 entries among the eight entries of the two initial tables.

Write-1 entries	Initial pairs	Both all-0	Both all-1
0	65,536	98.81%	0.00%
1	524,288	87.48%	8.30%
2	1,835,008	75.11%	16.99%
3	3,670,016	62.04%	26.80%
4	4,587,520	48.84%	38.83%
5	3,670,016	36.33%	53.08%
6	1,835,008	24.41%	68.60%
7	524,288	12.64%	84.05%
8	65,536	0.00%	99.40%

Perpetual rewriting. The 358,530 pairs that never freeze reach 111,890 cycles with periods between 4 and 96. In 286,646 of these pairs only one of the two programs keeps changing while the other is frozen. Period 4 alone accounts for 228,352 of the pairs.

E.5 Interpretation and scope

Why the uniform tapes dominate. Under the layout (79) the stored write bit is the symbol written. A machine that mostly writes 0 therefore fills its partner's tape with 0s, which turns the partner's write bits, move bits, and state bits toward the all-0 table, and the partner then writes 0s back. The uniform tapes are the fixed points of this feedback, and Table 3 is consistent with a competition between the two symbols decided largely by the initial write bits.

This account is an interpretation of the enumeration. We have not derived the basin sizes, the bias toward the all-0 pair, or the conditional probabilities in Table 3.

Dependence on the encoding. The self-reproducing property of the uniform tapes is a property of the encoding, not only of the machines. If a stored 0 in a write cell meant "write 1", the all-0 tape would no longer reproduce itself. The ordering of the three fields within an entry and of the four entries along the tape also fixes which cells a moving head meets, and we have not found a symmetry of the model that exchanges the all-0 and all-1 tapes. No other encoding has been run, so the present results do not establish which features are generic.

Other limits. The enumeration covers $n = 2$ states, two symbols, a 12-cell tape, and exactly two machines. The head-start conventions cover a small part of the 576 possible head configurations per pair, so tables with $\omega = 0$ or $\sigma = 0$ in the reference ensemble could appear or persist from other head starts. Larger tables require longer tapes and cannot be enumerated exhaustively in the same way.

Reproduction. The implementation is `turing_machines/mutual_rewriting.py`, run as `python -m scripts.mutual_rewriting`; options select the initial head cells and states or a random sample of pairs. The reference run is archived in `results/mutual_rewriting/`, with per-table values of ω, σ, and c in `machines.csv`, every attractor in `attractors.npz`, and the transient, period, and attractor of every initial pair in `pairs.npz`. Table identifiers there follow the labeling used by `alignment_census.py`. The update rule is tested against a scalar implementation written from the equations above, and the transients and periods against brute-force cycle detection, in `tests/test_mutual_rewriting.py`.

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